

EduTrack

Industry-Grade Deployment Checklist

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School Management System · Android · AI Assistant · Multi-Language

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■ Done

■ ■ Partial / In Progress

■ Not Started

CURRENT STATUS SUMMARY

Frontend	■ Done
Backend	■ Done
Android APK	■ Done — runs on device
Database	■■ Needs Supabase migration + RLS
Security	■■ Needs hardening
Testing	■ Not started
AI Assistant (Sage/Zeni)	■ Not started
Language Feature	■ Not started
Play Store	■ Not started
Legal/Compliance	■ Not started
Monitoring	■ Not started
Branding/Design	■■ Partially done

1. SECURITY

1.1 Supabase / PostgreSQL Database Security

- Enable Row Level Security (RLS) on every table
- Configure RLS policies e.g. `auth.uid() = user_id`
- Never expose the service role key in the frontend — use anon key only
- Use Supabase Auth for authentication
- Enable SSL on all database connections
- Set up automated daily database backups
- Never commit `.env` file to GitHub

1.2 Advanced Security

- JWT token expiry and refresh strategy
- SQL injection protection
- XSS (Cross-Site Scripting) protection
- CSRF protection
- API key rotation policy
- Data encryption at rest and in transit
- Secure session management
- Brute force protection — lock account after failed logins
- Two-factor authentication (2FA)
- Input validation and sanitization on all forms
- Rate limiting on all API endpoints
- HTTPS enforced everywhere

2. MOBILE APP

2.1 Core Mobile

- Android APK builds successfully
- App runs on Android phone
- App icon and splash screen designed
- App tested on multiple screen sizes
- iOS version built (requires a Mac or cloud build service)
- Release APK signed with keystore for Play Store

2.2 Advanced Mobile Features

- Push notifications (assignments, grades, announcements)
- Biometric login (fingerprint / face ID)
- Deep linking support
- App versioning strategy defined
- Handle no internet connection gracefully (offline mode)
- Background sync when connection restores
- Battery and data usage optimisation
- Auto-update mechanism

3. TESTING

3.1 Testing Requirements

- Unit tests for every function
- Integration tests for all API endpoints
- End-to-end testing
- User Acceptance Testing (UAT) with real students
- Performance and load testing
- Security penetration testing
- Cross-browser testing
- Cross-device testing (different Android versions)
- Error handling tested for all edge cases

4. AI STUDENT ASSISTANT (Sage / Zeni)

4.1 AI Assistant Features

- Name decided — suggested: Sage, Zeni, Nova, Lumio
- Integrated with AI API (Groq recommended — already familiar)
- Personality and tone defined — friendly, student-focused
- Context-aware — knows student's subjects and grades
- Answers curriculum-related questions
- Study plan generator
- Quiz generator from student notes
- Assignment helper (guides without doing it for them)
- Progress tracker with AI-generated insights
- Conversation history saved per student
- Content moderation to prevent misuse
- Fallback message when AI API is unavailable

5. MULTI-LANGUAGE FEATURE

5.1 Language Support

- i18next library integrated into frontend
- English supported (default)
- Zulu translation added
- Sepedi translation added (developer's home language)
- Afrikaans translation added
- Sotho translation added
- Auto-detect phone language on first launch
- Manual language switcher in app settings
- All error messages translated
- All push notifications translated
- Date and number formats per locale

6. BACKEND & DATABASE

6.1 Backend Requirements

- Node.js/Express backend built and running
- Migrate database to Supabase/PostgreSQL

- All API endpoints documented (Postman/Swagger)
- Environment separation: dev, staging, production
- API versioning strategy (e.g. /api/v1/)
- Error handling middleware on all routes
- Request logging implemented
- Database connection pooling configured
- Database indexes on frequently queried columns

7. ARCHITECTURE & CODE QUALITY

7.1 Code Standards

- Proper folder structure maintained
- Code documentation and comments
- ESLint and Prettier configured
- Git branching strategy: main, develop, feature branches
- CI/CD pipeline set up (auto deploy on GitHub push)
- Docker containerisation
- Environment variables documented (not values, just keys)

8. USER MANAGEMENT

8.1 User Features

- User roles defined: Student, Teacher, Admin, Parent
- Account deletion (POPIA compliance)
- Password reset via email
- Profile management (edit name, photo etc)
- Profile picture / avatar upload
- Account suspension system for admins
- Parent portal to monitor student progress

9. MONITORING & ANALYTICS

9.1 Monitoring Tools

- Error tracking with Sentry (free tier available)
- API response time monitoring
- Database query performance monitoring
- Crash reporting on mobile app
- User analytics — how people use the app
- User feedback system built into the app
- Uptime monitoring (UptimeRobot — free)

10. DESIGN & BRANDING

10.1 Design Requirements

- Consistent design system: colours, fonts, spacing
- App logo professionally designed
- Onboarding screens for new users

- Empty states designed (what shows when no data)
- Error pages designed (404, 500 etc)
- Loading states and skeleton screens
- Mobile responsive layout optimised
- Accessibility: colour contrast, font sizes (WCAG 2.1)
- Dark mode support

11. LEGAL & COMPLIANCE

11.1 Legal Requirements

- Privacy Policy page published
- Terms and Conditions page published
- POPIA compliance (South African data protection law)
- Cookie policy if applicable
- Data retention policy defined
- Parental consent flow (app targets under 18s)
- Accessibility compliance (WCAG 2.1)

12. GOOGLE PLAY STORE

12.1 Play Store Requirements

- Google Play Developer account created (\$25 once-off fee)
- Release APK signed with a keystore file
- App icon created — 512x512px PNG
- Feature graphic created — 1024x500px
- Minimum 4 screenshots per device type
- Short description written (max 80 characters)
- Full description written (max 4000 characters)
- Privacy policy URL provided
- Content rating questionnaire completed
- Target audience defined
- App category selected
- Pricing set (free or paid)
- Countries selected for distribution
- App tested on multiple devices before submission

13. BUSINESS & GROWTH

13.1 Business Strategy

- Monetisation strategy defined (freemium, school subscriptions, ads?)
- Competitor analysis completed
- Target market clearly defined (SA schools, townships, private?)
- Support email and helpdesk set up
- App Store Optimisation (ASO) — keywords and description optimised
- Social media presence created for EduTrack
- School outreach plan (pilot schools identified)
- User retention strategy planned

14. RECOMMENDED PRIORITY ORDER

1	Supabase Migration + RLS Security	Everything depends on a secure database
2	App Icon + Splash Screen	Makes the app look professional immediately
3	AI Assistant (Sage/Zeni) — Groq API	Biggest differentiating feature
4	Multi-Language Feature — i18next	Unique SA value proposition
5	Push Notifications	Critical for student engagement
6	Testing (Unit + Integration)	Required before Play Store submission
7	Privacy Policy + Terms + POPIA	Legal requirement for Play Store
8	Signed Release APK	Play Store requires signed build
9	Google Play Store Submission	Final deployment milestone
10	Monitoring + Analytics	Track real-world usage post-launch

You built a working full-stack app, converted it to Android, and got it running on your phone — solo, as a final-year student. That is genuinely impressive. This checklist is your roadmap to taking EduTrack from a great student project to an industry-grade product. Take it one step at a time. ■